DRUGS · HEAD AND NECK ONCOLOGY

Carboplatin

Carboplatin is a platinum compound with the same DNA cross-linking action as cisplatin but a different toxicity profile: much less nephrotoxicity, neurotoxicity, ototoxicity and vomiting, and more myelosuppression, especially thrombocytopenia. In head and neck cancer it is used for patients who cannot receive cisplatin, in palliative combinations and in some induction regimens. Its dose is calculated from renal function.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

CARBOPLATIN

	SUMMARY
Class	Platinum compound
ENT uses	Palliative chemotherapy for recurrent or metastatic squamous cell carcinoma (with 5-FU, a taxane, or immunotherapy as in KEYNOTE-048); concurrent chemoradiotherapy in patients unfit for cisplatin (less evidence); some nasopharyngeal and sinonasal regimens
Dose	By the Calvert formula: dose (mg) = target AUC x (GFR + 25), commonly AUC 5 every 3 weeks, or lower weekly doses with radiotherapy
Toxicity	Myelosuppression (platelets most), less renal, nerve and ear toxicity than cisplatin

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Choosing carboplatin

When renal function is poor, hearing is already impaired, neuropathy is present or the patient is frail, carboplatin is substituted for cisplatin, accepting probably lower efficacy with radiotherapy. Ototoxicity is uncommon at standard doses but occurs with high doses and in children.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Thrombocytopenia, neutropenia and anaemia (nadir at 2 to 3 weeks); nausea; hypersensitivity after repeated cycles; hypomagnesaemia.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- How its toxicity differs from cisplatin.
- The Calvert formula.

4.2 Common pitfalls

COMMON PITFALLS

- Assuming carboplatin carries no ototoxic risk.

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